Lección 5

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Plan de trabajo

Teoría de relatividad

– Cinemática Relativista:

Dispersión de partículas. Invariantes de Mandelstam

– Aplicaciones: ... efecto de Doppler relativista

Formulación Lagrangiana

– Repaso de conceptos de pregrado (mini-quiz)

... Variables generalizadas

... Clasificación de Sistemas Mecánicos

... Principio de mínima acción y variables generalizadas

► **Nuestro Syllabus** (**FIS-310**):

UNIDAD 1. Teoría de la Relatividad Especial
1.1. Transformaciones de Galileo
1.2. Experimento de Michelson y Morley
1.3. Postulados de Einstein
1.4. Transformaciones de Lorentz, Grupo de Lorentz
1.5. Espacio-tiempo de Minkowski y vectores 4-dimensionales
1.6. Cinemática Relativista
1.7. Dinámica Relativista
1.8. Aplicaciones: la paradoja de los gemelos, efecto Doppler relativista

UNIDAD 2. Formulación Lagrangiana
2.1. Variables Generalizadas
2.2. Clasificación de los Sistemas Mecánicos
2.3. Elementos de Cálculo Variacional
2.4. Ecuaciones de Lagrange
2.5. Aplicaciones: movimiento en un campo central, observador inerciales

FIS-345:

Unidad I: Principios Variacionales, Lagrangianos y Hamiltonianos
- Principio de Hamilton y Lagrangianos
- Simetrías y Leyes de Conservación (Teorema de Noether)

Unidad II: Algunas Aplicaciones
- Movimiento en campo de Fuerzas Centrales
- Movimiento de Sólido Rígido

FIS-426:

1) Principios Variacionales, Lagrangianos y Hamiltonianos
- Principio de Hamilton y Lagrangianos
- Hamiltonianos
- Otros principios variacionales
- Simetrías y Leyes de Conservación (Teorema de Noether)

Recap: Conservation of momentum

- Energy-momentum conservation in collision of free particles:

– Require energy and momentum conservation

$$\sum_i E_i^{(\text{before})} = \sum_j E_j^{(\text{after})}, \qquad \sum_i \boldsymbol{p}_i^{(\text{before})} = \sum_j \boldsymbol{p}_j^{(\text{after})}, \qquad E_i = c\sqrt{\boldsymbol{p}_i^2 + m_i^2 c^2}$$

where E_i is the energy of particle $i = 1, 2, \ldots$ with mass m_i

* Algebraic system, can solve and get relations between energy/momenta/angles (tedious due to numerous $\sqrt{(\ldots.)}$ from energy coservation)

- **Energy-momentum conservation in covariant form**:

$$\sum_i p_i^{\mu\ (\text{before})} = \sum_j p_j^{\mu\ (\text{after})}$$

e.g. for elastic $1 + 2 \to 1' + 2'$ scattering:

$$p_{1i}^\mu + p_{2i}^\mu = p_{1f}^\mu + p_{2f}^\mu, \qquad p_a \cdot p_a = m_a^2 \quad (p_a \cdot p_b \equiv \eta_{\mu\nu} p_a^\mu p_b^\nu) \tag{1}$$

* Consider all components of p^μ as "independent" and impose $p_a^2 \equiv p_a \cdot p_a = m_a^2$ later
* Can rewrite (1) in many equivalent forms, e.g:

$$p_{1i}^\mu - p_{1f}^\mu = p_{2f}^\mu - p_{2i}^\mu, \qquad p_{1i}^\mu - p_{1f}^\mu + p_{2i}^\mu = p_{2f}^\mu,, \qquad p_{1i}^\mu - p_{1f}^\mu + p_{2i}^\mu - p_{2f}^\mu = 0. \ \ldots, \tag{2}$$

which after squaring give nontrivial relations between scalar products $p_a \cdot p_b$

Relativistic elastic scattering

- Kinematics of initial particles is known, final state: unknown, but bound by energy-momentum conservation
- Numerous relations between scalar products $\Rightarrow$

$$\underbrace{4 \times 2}_{\text{components}} - \underbrace{4}_{\text{energy}-\text{mom. conservation}} - \underbrace{2}_{p_f^2 = m_f^2 c^2} = 2$$

  independent variables (e.g. energy and angle of one particle); all others are fixed
- Our goal: Lorentz-covariant formulation for observables (e.g. cross-sections)

$$p_{1i} \cdot p_{2i} = p_{1f} \cdot p_{2f} \quad (2)$$

$$m_1^2 - p_{1f} \cdot p_{1i} = m_2^2 - p_{2f} \cdot p_{2i} \quad (3)$$

$$p_{1f} \cdot p_{2i} = p_{2f} \cdot p_{1i} \quad (4)$$

$$p_{1i} \cdot p_{2i} + m_1^2 = p_{1i} \cdot p_{1f} + p_{2i} \cdot p_{1f} \quad (5)$$

$$p_{1f} \cdot p_{2f} + m_1^2 = p_{1i} \cdot p_{1f} + p_{1i} \cdot p_{2f} \quad (6)$$

$$p_{1i} \cdot p_{2i} + m_2^2 = p_{2i} \cdot p_{2f} + p_{1i} \cdot p_{2f} \quad (7)$$

Mandelstam invariants - Elastic scattering

Previous results:

- Numerous relations between scalar products $\Rightarrow 4\times 2-4-2=2$ independent variables (e.g. energy and angle of one particle); all others are fixed
- Our goal: Lorentz-covariant formulation for observables (e.g. cross-sections)

Mandelstam invariants: Lorentz-invariant variables defined as

$$s := (p_{1i} + p_{2i})^2 = (p_{1f} + p_{2f})^2$$
$$t := (p_{1f} - p_{1i})^2 = (p_{2f} - p_{2i})^2$$
$$u := (p_{1f} - p_{2i})^2 = (p_{2f} - p_{1i})^2$$

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^3 \quad (A)$$

Straightforward relation to energies, scattering angles in various frames

- Usually s is called invariant collision energy
  - In the frame where the particle 2 (target) is at rest, $s = 2E_1 m_2 + m_1^2 + m_2^2$
  - In the C-frame (both particles move in opposite directions with $\boldsymbol{p}_{1i} = -\boldsymbol{p}_{2i}$) can get $s = (E_{1i} + E_{2i})^2$
- Usually t is called invariant momentum transfer; for <u>elastic</u> scattering:
  - In the frame where the particle 2 is at rest, $t = -2m_2 (E_{1i} - E_{1f}) \leq 0$
  - In the C-frame $t = -2|\boldsymbol{p}|^2 (1 - \cos\theta)$

Mandelstam variables-Graphic illustration

Assume that we have some inelastic process

$$1 + 2 \to 3 + 4.$$

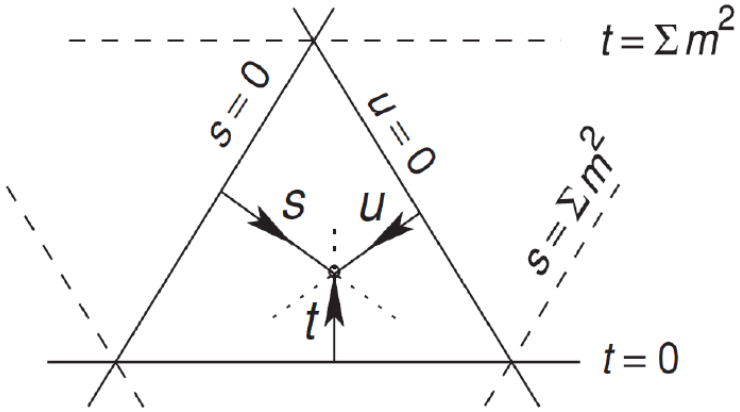
Masses of all particles 1,2,3,4 are different. Then Mandelstam variables s, t, u are bound by

$$s + t + u = m_1^2 + m_2^2 + m_3^2 + m_4^3 \quad (A)$$

▷Relation (A) implies that only 2 variables are independent

▷ Viviani's theorem: for any point inside equilateral triangle sum of distances equals height of triangle

⇒Graphical illustration of relation (A):



Mandelstam variables-Exercises

Assume that we study process

$$1 + 2 \to 3 + 4.$$

Masses of all particles 1,2,3,4 are different. Prove that the Mandelstam variables s, t, u are bound by inequalities

$$\begin{aligned}
s &= (p_1 + p_2)^2 \geq (m_1 + m_2)^2, \quad (A) \\
s &= (p_3 + p_4)^2 \geq (m_3 + m_4)^2, \quad (B) \\
t &= (p_2 - p_4)^2 \leq (m_2 - m_4)^2, \quad (C) \\
t &= (p_1 - p_3)^2 \leq (m_1 - m_3)^2, \quad (D) \\
u &= (p_1 - p_4)^2 \leq (m_1 - m_4)^2, \quad (E) \\
u &= (p_2 - p_3)^2 \leq (m_2 - m_3)^2. \quad (F)
\end{aligned}$$

Help: Prove first that $p_a \cdot p_b \geq m_a m_b$ where $p_a^2 = m_a^2, p_b^2 = m_b^2$. Since this is Lorentz-invariant, may use frame where one of the particles is at rest.

Mandelstam variables-Graphic illustration

Assume for simplicity that scattering is elastic:
– all masses are equal, $m_i \equiv m$
– work in CM frame of particles $1, 2$ (C-frame), so

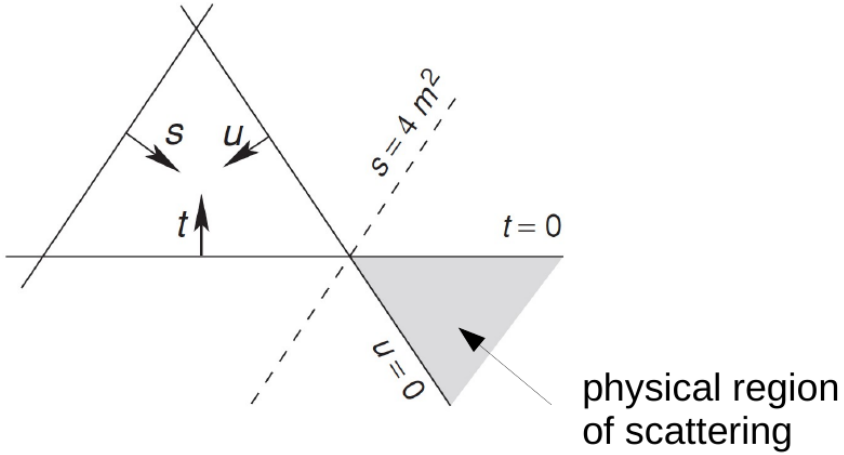
$$p_1 = (E, \boldsymbol{p}), \qquad p_2 = (E, -\boldsymbol{p}), \quad E = \sqrt{m^2 + \boldsymbol{p}^2}$$

Mandelstam variables:

$$s = (p_1 + p_2)^2 = 2m^2 + 2\left(E^2 + \boldsymbol{p}^2\right) = 4m^2 + 4\boldsymbol{p}^2 \geq 4m^2$$

After elastic scattering the energy is conserved, only directions of $\boldsymbol{p}_i$ change:

$$t = (p_{1f} - p_{1i})^2 = \underbrace{(E_{1f} - E_{1i})^2}_{0} - (\boldsymbol{p}_{1f} - \boldsymbol{p}_{1i})^2 = -2|\boldsymbol{p}|^2 (1 - \cos\theta) \leq 0$$



Mandelstam variables-Exercises

Assume we study process

$$1 + 2 \to 3 + 4.$$

and we've chosen 3 independent momenta q_1, q_2, q_3 (the 4th might be found from energy-momentum conservation) and construct vector

$$L_\alpha = \varepsilon_{\alpha\beta\mu\nu} q_1^\beta q_2^\mu q_3^\nu .$$

Show that in the rest frame of particle $1, 2$ or 3 the vector L_α cannot have 0-component, and for this reason

$$L_\alpha L^\alpha \equiv -\boldsymbol{L}^2 \leq 0 \quad (A)$$

Show that in arbitary frame

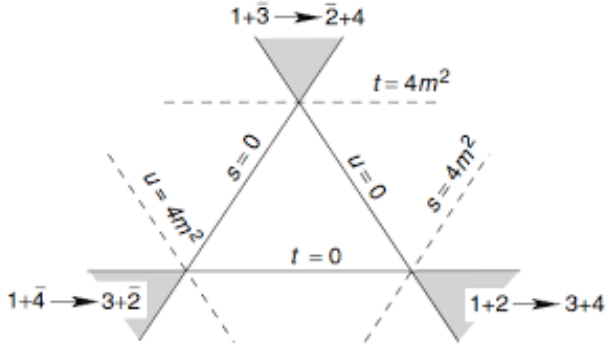
$$-L_\alpha L^\alpha = \det \begin{vmatrix} q_1^2 & q_1 \cdot q_2 & q_1 \cdot q_3 \\ q_1 \cdot q_2 & q_2^2 & q_2 \cdot q_3 \\ q_1 \cdot q_3 & q_2 \cdot q_3 & q_3^2 \end{vmatrix} = \det \begin{vmatrix} m_1^2 & \frac{s - m_1^2 - m_2^2}{2} & \frac{m_1^2 + m_3^2 - t}{2} \\ \frac{s - m_1^2 - m_2^2}{2} & m_2^2 & \frac{m_2^2 + m_3^2 - u}{2} \\ \frac{m_1^2 + m_3^2 - t}{2} & \frac{m_2^2 + m_3^2 - u}{2} & m_3^2 \end{vmatrix} \geq 0$$

-this gives nontrivial conditions for possible values of s, t, u (*Kibble*)

Mandelstam variables

Further reading:

- *Problems* in Landau, Vol. 2, Section 13
- Vol. 4 of Berestetskii, V.B., Lifshitz, E.M. and Pitaevskii, L.P. (1974) Relativistic Quantum Theory. Pergamon Press, Section 66-67
- If You will try to seek Mandelstam variables on your own, You may find something like:



– in Quantum Field Theory it is shown that the same cross-section (function) describes the process $1 + 2 \to 3 + 4$, as well as processes which might be obtained replacing particle with momentum p in initial state with its antiparticle with momentum $-p$ in final state (and vice versa). This leads to permutation of variables s, t, u and explains why we get into "unphysical" region (In the above-given plot $\bar{a}$ implies antiparticle of particle a).

Relativistic Doppler effect

We've discussed several examples how the relativistic effects manifest themselves, and in your homeworks ("Tareas") you will analyze more complicated systems

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There is just one more topic - "Doppler effect" which we need to discuss and which is mentioned in our syllabus, so need to discuss it

– Please try to recall from undergraduate courses what is Doppler effect ?
  * Should we move with relativistic velocities to see it ?

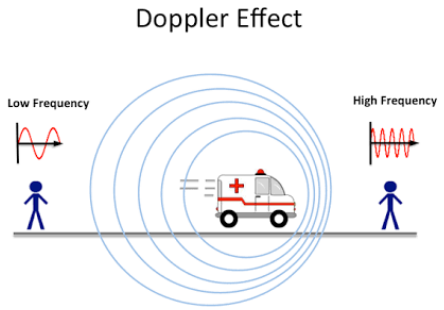
Relativistic Doppler effect

We've discussed several examples how the relativistic effects manifest themselves, and in your homeworks ("Tareas") you will analyze more complicated systems

There is just one more effect which we need to discuss and which is mentioned in our syllabus, so need to discuss it

- The Doppler effect is a change of observed frequency when the observer and the source are moving inside the medium

►Exists in classical mechanics (e.g. acoustics):



$$\omega_{\text{observer}} = \omega_{\text{source}} \left(\frac{1 - \frac{\boldsymbol{v}_o \cdot \boldsymbol{n}}{u}}{1 - \frac{\boldsymbol{v}_s \cdot \boldsymbol{n}}{u}} \right)$$

▷$\vec{\boldsymbol{v}}_s$ and $\vec{\boldsymbol{v}}_o$ are velocities of the source and the observer with respect to the medium

▷ $\boldsymbol{n}$ is the direction from source to observer, $n^2 = 1$

▷ u is the speed of wave in the medium (e.g. acoustic wave)

►This effect is NOT related to Lorentz time dilation!

Relativistic Doppler effect

- In electromagnetism we do not have "medium" (ether), all reference frames are equivalent, so only the relative motion matters.

►Doppler effect: change of frequency (wavelength) due to relative motion of the source/detector.

–Similar to nonrelativistic case, have linear term

$$\left(\frac{\Delta\omega}{\omega}\right)_{\text{nonrel}} \sim \frac{|v_i|}{c}$$

(due to motion, change time between arrival of the surfaces of constant phase)

–Additional contribution: Lorentz time dilation effects,

$$\Delta t_{\text{lab}} = \frac{\Delta t_{\text{r.f.}}}{\sqrt{1 - v^2/c^2}} \approx \Delta t_{\text{r.f.}} \left(1 + \frac{v^2}{2c^2}\right) \Rightarrow \left(\frac{\Delta\omega}{\omega}\right)_{\text{rel}} \sim \frac{v^2}{2c^2} \ll \left(\frac{\Delta\omega}{\omega}\right)_{\text{nonrel}}$$

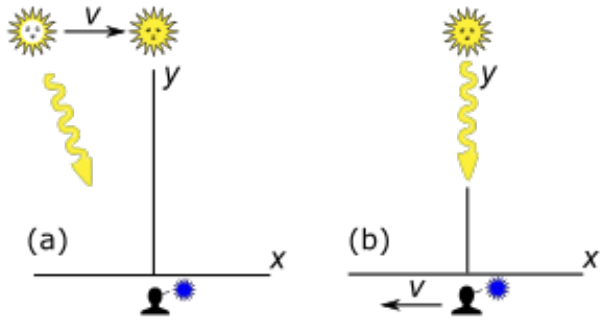
become significant when velocities $|v_i| \sim c$

►**Question:**

If relativistic Doppler is so tiny, how can we observe it/distinguish from usual (nonrelativistic)?

How to measure tiny relativistic Doppler?

- Can observe Doppler effect analyzing the shift of the spectral lines (measurable with exceptionally good precision, sensitive test of relativistic effects)
  - Nonrelativistic Doppler depends on orientation of velocity $\boldsymbol{v}$: depends on projection of $\boldsymbol{v}$ onto direction of observation
  - Relativistic Doppler does not depend on orientation of velocity $\boldsymbol{v}$



$\Rightarrow$In (a) dominates linear nonrelativistic Doppler

$\Rightarrow$In (b) linear nonrelativistic Doppler vanishes, can see relativistic Doppler

Relativistic Doppler effect (mathematics)

Let's assume that a source emits monochromatic plane wave in its rest frame:

$$\begin{pmatrix} \boldsymbol{E} \\ \boldsymbol{B} \end{pmatrix} = \begin{pmatrix} \boldsymbol{E}_0 \\ \boldsymbol{B}_0 \end{pmatrix} e^{i(\omega t - \boldsymbol{k}\cdot\boldsymbol{x})}, \quad \omega = |\boldsymbol{k}|\ c,$$

$$(\omega t - \boldsymbol{k}\cdot\boldsymbol{x}) = k^\mu x_\mu, \quad k^\mu = \left(\frac{\omega}{c},\ \boldsymbol{k}\right).$$

– For a moment ignore $\boldsymbol{E}_0, \boldsymbol{B}_0$ which determine polarization
– Plane wave maintains its form in any reference frame, so k^μ MUST transform as components of 4-vector

$$\omega' = \frac{\omega - V\,k_x}{\sqrt{1 - V^2/c^2}}, \quad k'_x = \frac{k_x - V\,\omega/c^2}{\sqrt{1 - V^2/c^2}}, \quad k'_y = k_y, \quad k'_z = k_z,$$

$$\omega = \frac{\omega' + V\,k'_x}{\sqrt{1 - V^2/c^2}}, \quad k_x = \frac{k_x + V\,\omega/c^2}{\sqrt{1 - V^2/c^2}}, \quad k'_y = k_y, \quad k'_z = k_z.$$

* Can see that even for $k'_x = 0$ (emission in transverse direction) the wave changes its frequency:

$$\omega = \frac{\omega'}{\sqrt{1 - V^2/c^2}} \approx \omega' \left(1 + \frac{V^2}{c^2}\right) \geq \omega'$$

This is manifestation of 2nd order Doppler effect due to relativistic correction

Relativistic Doppler effect (mathematics)

Generalization of our result for arbitrary direction of $\boldsymbol{v}$

$$k_0' = \gamma\left(k_0 - \frac{\boldsymbol{v}\cdot\boldsymbol{k}}{c}\right),$$
$$k_{||}' = \gamma\left(k_{||} - \frac{v}{c}k_0\right),\ \boldsymbol{k}_\perp' = \boldsymbol{k}_\perp$$

Recall that $\omega = c|\boldsymbol{k}|$, so can get for transformation of frequencies (Doppler effect) and directions of light rays

$$\Rightarrow \omega' = \gamma\omega\left(1 - \frac{v}{c}\cos\theta\right),$$
$$\Rightarrow \tan\theta' = \frac{\sin\theta}{\gamma(\cos\theta - v/c)},$$

(θ, θ' are measured w.r.t. direction of $\boldsymbol{v}$)

– If monochromatic source with frequency ω emits isotropically in its own reference frame, in any other frame observed frequency will depend on direction (angle θ)

– Note that for $\theta = \pi/2$ get $\omega_\perp = \omega\gamma > \omega$, transverse Doppler shift (merely relativistic effect, $\sim v^2/c^2$).

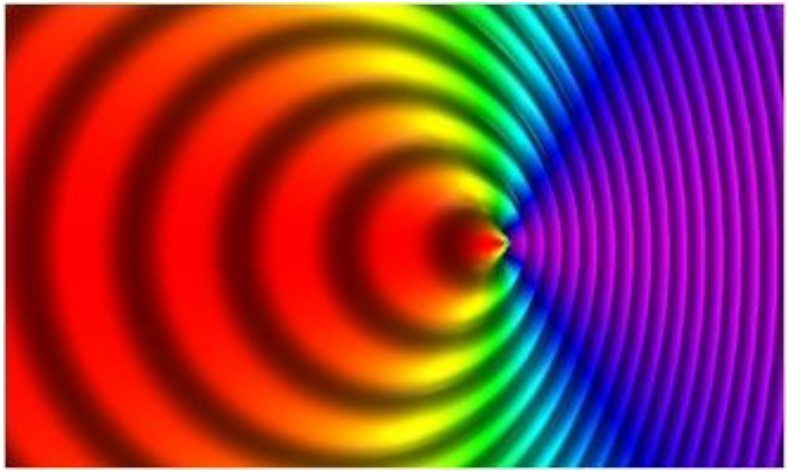
Relativistic Doppler effect (mathematics)

– If monochromatic source with frequency ω' emits isotropically in its own reference frame, in any other frame observed frequency will depend on direction (angle θ)

$$\Rightarrow \omega = \frac{\omega' \sqrt{1 - v^2/c^2}}{\left(1 - \frac{v}{c}\cos\theta\right)}$$

As the source passes by, the angle θ and frequency ω change constantly from $\omega > \omega'$ at $\theta \ll \pi/2$ (blushift) up to $\omega < \omega'$ for $\theta \gg \pi/2$ (redshift)

Nice visualization from Wikipedia:



► (Green) source moves to the right

– Observer sees the spectrum blue-shifted, if the source approaches to him

– Observer sees the spectrum red-shifted, if the source moves away from him.

Block 1 - Closing remark

We've covered all the topics of block 1.

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Please let me know if You have any questions?

Formalismo de Lagrange - syllabus

- **FIS-310**:

  UNIDAD 2. Formulación Lagrangiana
  2.1. Variables Generalizadas
  2.2. Clasificación de los Sistemas Mecánicos
  2.3. Elementos de Cálculo Variacional
  2.4. Ecuaciones de Lagrange
  2.5. Aplicaciones: movimiento en un campo central, observadores no inerciales

- **FIS-345**:

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  - Simetrías y Leyes de Conservación (Teorema de Noether)

  Unidad II: Algunas Aplicaciones
  - Movimiento en campo de Fuerzas Centrales
  - Movimiento de Sólido Rígido

- **FIS-426**:

  1) Principios Variacionales, Lagrangianos y Hamiltonianos
  - Principio de Hamilton y Lagrangianos
  - Hamiltonianos
  - Otros principios variacionales
  - Simetrías y Leyes de Conservación (Teorema de Noether)

Lagrangian formalism

So far we discussed kinematics and energy-momentum conservation.
−Now need to consider dynamics in detail

- We will start with nonrelativistic Lagrangian formulation and demonstrate its equivalence to the second law of Newton

▶ *It is expected that You've seen the formalism earlier in undergraduate courses. For this reason we'll start with short list of questions (micro-exam for diagnostics). The goal is to see what You remember. Please be active and reply questions if You can (or correct/complement replies of other students). Later we'll discuss these topics, passing some basic topics in "review" mode and paying more attention to topics which caused difficulties.*

Control question 1

Please formulate the minimal action principle and explain how to construct the action for a pointlike particle in external field.

Control question 2

Assume that we have a system made of a star and 2 planets. How many degrees of freedom has this system ? How many lagrangians ?

Control question 3

What is more general, Lagrangian or Newtonian formulation of mechanics?

– Please give examples of systems which might be described in one approach but not in the other.

Control question 3.1

We discussed explicitly Lagrangian or Newtonian formulations ...

–Do You know any other formulations of mechanics, apart from these two ?

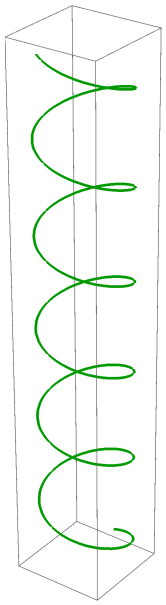
Control question 4

Is lagrangian uniquely defined?
Can we have two lagrangians which lead to the same equation of motion ?

Control question 5

Is lagrangian invariant under Galileo or Lorentz transforms ?
How this affects covariance of the equations of motion ?

Control question 6



Assume that a charged particle moves in the external potential, which in cylindric coordinates is given by $U = U(r, z - a\phi)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon, \; \phi \to \phi + \frac{\epsilon}{a} \quad (1)$$

What can we say about **integrals of motion** of the system in this field ?

Assume that a charged particle moves in the external potential, which in cylindric coordinates is given by $U = U(z - a\,r)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon, \; r \to r + \frac{\epsilon}{a} \quad (2)$$

What can we say about **integrals of motion** of the system in this field ? Explain your result

Control question 7

Assume that we study a particle which moves in central field $U(r)$.
–In which reference frame should we write the Euler-Lagrange equations:
Cartesian, spherical, cylindrical ... ???

Control question 8

Can we use a lagrangian approach in noninertial reference frame?
−Please explain in detail ...

Control question 9

Assume that we have a system of relativistic particles.

– Can we apply the Lagrangian formalism for its description?
  Please explain in detail ...

Lagrangian formalism

- Formulation in terms of Equations of Newton is useful, yet is not unique.

- Lagrangian formulation is based on Hamilton's minimal action principle

–... it is alternative to Newtonian formulation

- Equivalent to the 2nd law of Newton

Advantages of Lagrange formulation

–More convenient & easy in curvilinear coordinates, noninertial reference frame ...

–Easy to implement conservation laws

–Easy to implement constraints

Drawback of Lagrange formulation

–Applicable only for conservative forces ($\vec{F} = \nabla U$),

... not for dissipative forces (friction, ...)

... Sometimes You may <u>accidently</u> get equations of motion (Euler-Lagrange equations) which coincide with equation in presence of absorptive system (We'll see an example later), yet this property is not universal.

Newtonian formulation (summary)

The first law of Newton

If the net force (the vector sum of all forces acting on an object) is zero, then the velocity of the object is constant *in the inertial rest frame*.

The second law of Newton

$$\frac{d\vec{p}}{dt} = m\,\frac{d\vec{v}}{dt} = m\,\vec{a} = \vec{F}\,(\vec{r},\ \vec{v},\ t) \qquad (1)$$

The third law of Newton

All forces between two objects exist in equal magnitude and opposite direction: if one object A exerts a force F_A on a second object B, then B simultaneously exerts a force F_B on A, and $\vec{F}_A = -\vec{F}_B$.

The three laws of Newton describe all the dynamics of the mechanical system.

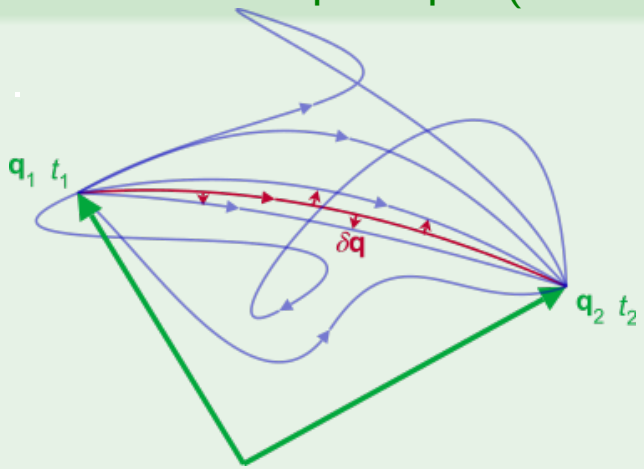
Laplace's determinism

The dynamics of the particle is fully determined by its initial position and velocities.

In some systems (and under additional conditions) there could be first integrals (also called "integrals of motion") which are constant along the trajectory. Typically, these integrals of motion are built of the quantities which contain only the first order derivatives of coordinates., so requirement that some quantity is constant is the 1st order ODE. Existence of such integrals may simplify the integration of the equations of motion.

Lagrangian formulation of mechanics

Minimal action principle (Hamilton)



For each dynamical system there is a functional called "action"

$$S = \int_{t_1}^{t_2} L\left(q_i,\, \dot{q}_k,\, t\right) dt$$

such that real trajectories correspond to the minimum of the action. The function L is Lagrangian of the system.

Euler-Lagrange equations

$$0 = \frac{\delta \mathcal{F}}{\delta q_a} = \frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right), \quad a = 1,\, \ldots,\, 3, \quad p_a \equiv \frac{\partial L}{\partial \dot{q}_a}$$

The quantity $\delta \mathcal{F}/\delta q$ is called **functional derivative,**

$$\frac{\delta \mathcal{F}}{\delta q_a(t)} := \frac{\partial L}{\partial q_a(t)} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) + \ldots + (-1)^n \frac{d^n}{dt^n}\left(\frac{\partial L}{\partial q_a^{(n)}}\right) + \ldots$$

Lagrangian formulation of mechanics

- The choice of Lagrangian

$$L = T - U = \frac{m\,\dot{\boldsymbol{q}}^2}{2} - U(\boldsymbol{q})$$

reproduces the second law of Newton:

$$\frac{\partial L}{\partial q_a} = -\frac{\partial U}{\partial q_a}, \quad \frac{\partial L}{\partial \dot{q}_a} = m\,\dot{q}_a, \quad \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) = m\,\ddot{q}_a$$

$$\Rightarrow m\ddot{\boldsymbol{q}} = -\boldsymbol{\nabla} U\left(\boldsymbol{q}\right)$$

Example

The lagrangian for the particle of mass m in the **homogeneous** gravitational field:

$$L = \frac{m\dot{\boldsymbol{r}}^2}{2} - mgz = \frac{m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right)}{2} - mgz \qquad (1)$$

- In general case (**non-homogeneous** field of the Earth) we should write:

$$L = \frac{m\dot{\boldsymbol{r}}^2}{2} + \frac{GMm}{r} = \frac{m\left(\dot{x}^2 + \dot{y}^2 + \dot{z}^2\right)}{2} + \frac{GMm}{\sqrt{x^2 + y^2 + z^2}} \qquad (2)$$

Nonuniqueness of the lagrangian

- The Lagrangian is not uniquely defined: multiplication of lagrangian by arbitrary **constant**,

$$L(q, \dot{q}, t) \to \lambda\, L(q, \dot{q}, t)\,, \qquad \lambda = \mathrm{const}$$

essentially does NOT affect the equations of motion (all terms are multiplied by the same const λ which eventually vanishes):

$$0 = \lambda \left[\frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a} \right) \right], \quad a = 1, \ldots, 3, \quad p_a \equiv \frac{\partial L}{\partial \dot{q}_a}$$

–Similarly, can check that adding a full derivative (d/dt) of *arbitrary* function $f(q, t)$

$$L \to L + \frac{d\, f(q,t)}{dt} = L + \frac{\partial f(q,t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(q,t)}{\partial q_a}$$

does not change the equations of motion and the trajectory.

Proof 1:

Minimal action principle (Hamilton)

$$S = \int_{t_1}^{t_2} L\left(q_i,\ \dot{q}_k,\ t\right) dt = \min$$

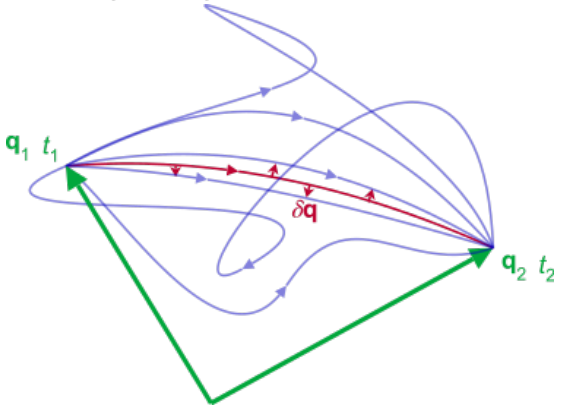
We can see that the transformation of Lagrangian

$$L \to L + \frac{d\,f(q,t)}{dt} = L + \frac{\partial f(q,t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(q,t)}{\partial q_a},$$

leads to modification of action

$$S \to S + \int_{t_1}^{t_2} dt\, \frac{d\,f(q,t)}{dt} = S + f(q_2, t_2) - f(q_1, t_1)$$

-while S changes, the change is sensitive only to endpoint values of f. The value of S does not change when we vary the curve at intermediate point, and thus modification of lagrangian cannot affect equation of motion.



Proof 2:

Let's show explicitly that Euler-Lagrange equations remain invariant under

$$L \to L + \frac{d\,f(q,t)}{dt} = L + \frac{\partial f(q,t)}{\partial t} + \sum_a \dot{q}_a \frac{\partial f(q,t)}{\partial q_a}$$

Change of different contributions in Euler-Lagrange equation:

$$\frac{\partial L}{\partial q_a} \to \frac{\partial L}{\partial q_a} + \frac{\partial^2 f(q,t)}{\partial q_a\,\partial t} + \sum_b \dot{q}_b \frac{\partial^2 f(q,t)}{\partial q_a \partial q_b}\,,$$

$$\frac{\partial L}{\partial \dot{q}_a} \to \frac{\partial L}{\partial \dot{q}_a} + \frac{\partial f(q,t)}{\partial q_a}\,,$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) \to \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) + \frac{d}{dt}\left(\frac{\partial f(q,t)}{\partial q_a}\right) = \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right) + \frac{\partial^2 f(q,t)}{\partial q_a\,\partial t} + \sum_b \dot{q}_b \frac{\partial^2 f(q,t)}{\partial q_a \partial q_b}$$

So we can see that in the difference

$$\frac{\partial L}{\partial q_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_a}\right),$$

all the red-colored terms will cancel, as expected.

Example 2: External EM field

Assume that You have a lagrangian

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} + e\,\dot{\boldsymbol{r}} \cdot \boldsymbol{A}(\boldsymbol{r}) - e\,\varphi(\boldsymbol{r})$$

where $\boldsymbol{A}, \varphi$ are some external fields.

– The "gauge transformation"

$$\varphi \to \varphi - \frac{\partial f}{\partial t}, \quad \boldsymbol{A} \to \boldsymbol{A} + \nabla f$$

where $f(\boldsymbol{r}, t)$ is arbitrary function, modifies the lagrangian as

$$L \to L + e\frac{\partial f}{\partial t} + e\dot{\boldsymbol{r}} \cdot \boldsymbol{\nabla} f = L + e\frac{df}{dt}$$

and thus does not change equations of motion

Lemma:

– This lagrangian corresponds to motion of a charged particle in external electromagnetic field.

$$\Rightarrow m\,\ddot{\boldsymbol{r}} = e\left(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}\right), \qquad \boldsymbol{B} = \boldsymbol{\nabla} \times \boldsymbol{A}, \quad \boldsymbol{E} = -\boldsymbol{\nabla}\varphi - \frac{\partial \boldsymbol{A}(\boldsymbol{r})}{\partial t}$$

Example 2: External EM field (proof)

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} + e\,\dot{\boldsymbol{r}}\cdot\boldsymbol{A}(\boldsymbol{r}) - e\,\varphi(\boldsymbol{r})$$

Generalized momentum:

$$p_a = \frac{\partial L}{\partial \dot{r}_a} = m\,\dot{\boldsymbol{r}} + e\,\boldsymbol{A}(\boldsymbol{r}) \neq m\,\dot{\boldsymbol{r}}$$

Equations of motion:

$$\frac{d\,\boldsymbol{p}}{dt} = m\,\ddot{\boldsymbol{r}} + e\,\frac{d\boldsymbol{A}(\boldsymbol{r})}{dt} = -e\partial_a\varphi(\boldsymbol{r}) + e\,\boldsymbol{\nabla}_A\,(\boldsymbol{v}\cdot\boldsymbol{A})$$

$$\frac{d\boldsymbol{A}(\boldsymbol{r})}{dt} \equiv \frac{\partial\boldsymbol{A}(\boldsymbol{r})}{\partial t} + (\boldsymbol{v}\cdot\nabla)\,\boldsymbol{A}(\boldsymbol{r})$$

$$\boldsymbol{v}\times[\boldsymbol{\nabla}\times\boldsymbol{A}] = \nabla_A\,(\boldsymbol{v}\cdot\boldsymbol{A}(\boldsymbol{r})) - (\boldsymbol{v}\cdot\nabla)\,\boldsymbol{A}(\boldsymbol{r})$$

$$\Rightarrow m\,\ddot{\boldsymbol{r}} = e\,(\boldsymbol{E} + \boldsymbol{v}\times\boldsymbol{B})\,, \qquad \boldsymbol{B} = \boldsymbol{\nabla}\times\boldsymbol{A}, \quad \boldsymbol{E} = -\boldsymbol{\nabla}\varphi - \frac{\partial\boldsymbol{A}(\boldsymbol{r})}{\partial t}$$

Lagrangian & Galilean transformations

Assume that a system of partciels is *closed*, *i.e.* particles interact only with each other, so the potential energy is given by $U = U(|\boldsymbol{q}_a - \boldsymbol{q}_b|)$. Find the transformation law of its Lagrangian under Galilean transformations. Demonstrate that the equations of Euler-Lagrange do not change their form (are "covariant").

Lagrangian & Galilean transformations

$$L = T - U = \sum_{a=1}^{3} \frac{m\,\dot{q}_a^2}{2} - U\left(q_a\right)$$

- The term with potential energy depends only on relative distances between particles, and since distances don't change under Galilean transformations, it remains invariant
- The term with kinetic energy is more challenging, since velocities transform as

$$\boldsymbol{v} \to \boldsymbol{v} + \boldsymbol{V},$$

so after squaring this result, we can see that the lagrangian is not invariant.

- Change of Lagrangian under Galilean transformation:

$$L \to L + \boldsymbol{V} \cdot \left(\sum_a m_a \boldsymbol{v}_a\right) + \left(\sum_a m_a\right) \frac{\boldsymbol{V}^2}{2} =$$

$$= L + \frac{d}{dt}\left[\boldsymbol{V} \cdot \left(\sum_a m_a \dot{\boldsymbol{q}}_a\right) + \left(\sum_a m_a\right) \frac{\boldsymbol{V}^2 t}{2}\right]$$

-the additional term is a full derivative, so it will not affect equations of motion.

Example 3

Find the equation of motion for a system with lagrangian given by

$$L = e^{-\gamma t}\left(\frac{m\,\dot{\boldsymbol{x}}^2}{2} - \frac{k\boldsymbol{x}^2}{2}\right)$$

Example 3 (solution)

$$L = e^{\gamma t}\left(\frac{m\,\dot{\boldsymbol{x}}^2}{2} - \frac{k\boldsymbol{x}^2}{2}\right)$$

$$\frac{\partial L}{\partial \boldsymbol{x}} = -k\,\boldsymbol{x}e^{-\gamma t}, \quad \frac{\partial L}{\partial \dot{\boldsymbol{x}}} = e^{\gamma t} m\dot{\boldsymbol{x}}$$

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\boldsymbol{x}}}\right) = \frac{d}{dt}\left(e^{\gamma t} m\dot{\boldsymbol{x}}\right) = \left(m\ddot{\boldsymbol{x}} + \gamma m\,\dot{\boldsymbol{x}}\right)e^{\gamma t}$$

$$\left(m\ddot{\boldsymbol{x}} + \gamma m\,\dot{\boldsymbol{x}}\right)\cancel{e^{-\gamma t}} = -k\,\boldsymbol{x}\cancel{e^{-\gamma t}}$$

$$m\ddot{\boldsymbol{x}} = -\gamma m\,\dot{\boldsymbol{x}} - k\boldsymbol{x}$$

So we got EOM in presence of velocity-dependent absorptive force (instead of $k\boldsymbol{x}^2/2$ could have any $U(\boldsymbol{x})$)

– This is exceptional case: if we have any other velocity dependence in absorptive force, or kinetic energy $T \neq mv^2/2$, this method won't work.

Control question

So far we considered lagrangians which include only $q(t)$ and $\dot{q}(t)$.

$-$Can the lagrangian include higher order derivatives, *e.g.* $\ddot{q}$, $\dddot{q}$?

$-$Can the lagrangian L include nonlocal terms, $e.g.$:

$$\dot{q}(t) \int d\tau \; K(t, \tau) \dot{q}(\tau)$$

where $K(t, \tau)$ is some kernel

Please explain in detail ...

Properties of Lagrangian

▷ For more then one particle generalization is

$$L = T - U = \sum_{a=1}^{3n} \frac{m\,\dot{q}_a^2}{2} - U\left(q_1, ..., q_{3n}\right) \qquad (2)$$

– Lagrangian (action) is defined only for the whole system, NOT for individual particles. It depends on all coordinates q_a and their derivatives $\dot{q}_a$.
– If the system includes 2 <u>non-interacting</u> parts, then its lagrangian is given by a sum

$$L = L_1 + L_2,$$

$$L_1 = \sum_{a=1}^{k} \frac{m\,\dot{q}_a^2}{2} - U_1\left(q_1, ..., q_k\right)$$

$$L_2 = \sum_{a=k+1}^{n} \frac{m\,\dot{q}_a^2}{2} - U_2\left(q_{k+1}, ..., q_n\right)$$

so the equations of motion (equations of Euler-Lagrange) for 2 subsystems decouple

Lagrangian formalism - curvilinear coordinates

- Earlier we saw that in Cartesian coordinates Euler-Lagrange equations coincide with Newtonian equations in Cartesian coordinates

$$m\ddot{\boldsymbol{q}} = -\boldsymbol{\nabla} U(\boldsymbol{q})$$

- For some problems (e.g. motion in spherically symmetric field) it might be more convenient to use other reference frame. However, in curvilinear coordinates consideration is tricky, since

$$\frac{d\boldsymbol{V}}{dt} = \sum_a \frac{d\left(V_a \boldsymbol{e}_a\right)}{dt} = \sum_a \frac{dV_a}{dt}\boldsymbol{e}_a + \sum_a V_a \frac{d\boldsymbol{e}_a}{dt}$$

  where $\boldsymbol{e}_a$are basis vectors, so if we will write Newtonian equations in components, we might get some "additional" terms.
- Now we are going to analyze how the equations of Euler-Lagrange look in other reference frames

Lagrangian formalism - curvilinear coordinates

Assume that now we make transformation of variables (e.g. from Cartesian system to spherical or cylindrical),

$$
\begin{aligned}
q_1 &= q_1\left(r_1,\, r_2,\, r_3\right), \\
q_2 &= q_2\left(r_1,\, r_2,\, r_3\right), \\
q_3 &= q_3\left(r_1,\, r_2,\, r_3\right).
\end{aligned}
$$

Rewrite the lagrangian in terms of variables r_i. Demonstrate that the Euler-Lagrange equations remain covariant (have the same form)

Transformation of variables:

$$q_a = q_a\left(r_1,\ r_2,\ r_3\right),$$

The lagrangian in terms of variables r_i:

$$\dot{q}_a = \sum_{b=1}^{3} \frac{\partial q_a}{\partial r_b}\,\dot{r}_b$$

$$L = T - U = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a\dot{r}_b}{2} - U\left(r_a\right), \quad \mu_{ab}(r) = m\sum_{k=1}^{3} \frac{\partial q_k}{\partial r_a}\,\frac{\partial q_k}{\partial r_b}$$

Example:
Transformation of variables (Cartesian→spherical):

$$q_x = r\sin\theta\cos\phi, \quad q_y = r\sin\theta\sin\phi, \quad q_z = r\cos\theta.$$

The lagrangian in terms of variables r, θ, ϕ.

$$L = T - U = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a\dot{r}_b}{2} - U\left(r_a\right)$$

$$T = m\frac{(\dot{r})^2 + r^2(\dot{\theta})^2 + r^2\sin^2\theta\,(\dot{\phi})^2}{2}$$

Technical detail: explicit evaluation of μ_{ab}

Transformation of variables:

$$q_x = r\sin\theta\cos\phi,$$
$$q_y = r\sin\theta\sin\phi,$$
$$q_z = r\cos\theta.$$

$$\mu_{ab}(r) = m\sum_{k=1}^{3}\frac{\partial q_k}{\partial r_a}\frac{\partial q_k}{\partial r_b}$$

$$\begin{aligned}\frac{\mu_{rr}}{m} &= \frac{\partial x}{\partial r}\frac{\partial x}{\partial r} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial r} + \frac{\partial z}{\partial r}\frac{\partial z}{\partial r}\\ &= \sin^2\theta\cos^2\phi + \sin^2\theta\sin^2\phi\\ &+ \cos^2\theta =\\ &= \sin^2\theta\underbrace{\left(\cos^2\phi + \sin^2\phi\right)}_{=1} +\\ &+ \cos^2\theta = 1\end{aligned}$$

$$\begin{aligned}\frac{\mu_{r\theta}}{m} &= \frac{\partial x}{\partial r}\frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial \theta} + \frac{\partial z}{\partial r}\frac{\partial z}{\partial \theta}\\ &= \sin\theta\cos\phi\, r\,\cos\theta\cos\phi +\\ &+ \sin\theta\sin\phi\, r\,\cos\theta\sin\phi\\ &- r\sin\theta\cos\theta = \ldots = 0\end{aligned}$$

$$\begin{aligned}\frac{\mu_{\theta\theta}}{m} &= \frac{\partial x}{\partial \theta}\frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial \theta}\frac{\partial y}{\partial \theta} + \frac{\partial z}{\partial \theta}\frac{\partial z}{\partial \theta}\\ &= (r\,\cos\theta\cos\phi)^2 +\\ &+ (r\,\cos\theta\sin\phi)^2 +\\ &+ (r\,\sin\theta)^2 = r^2\end{aligned}$$

etc.

Explicit expression for μ_{ab} in any curvilinear coordinates

We found that μ_{ab} has the following values:

–**Cartesian**:

$$\mu_{ab}^{(\mathrm{Cart})} = m\delta_{ab}$$

–**Spherical**:

$$\mu_{ab}^{(\mathrm{Sph.})} = \begin{pmatrix} m & & \\ & mr^2 & \\ & & mr^2 \sin^2\theta \end{pmatrix}$$

$$T = \frac{m\dot{r}^2 + mr^2\dot{\theta}^2 + mr^2 \sin^2\theta\dot{\varphi}^2}{2}$$

–If we repeat the same derivation in **cylidrical** coordinates, we'll get

$$\mu_{ab}^{(\mathrm{Cyl.})} = \begin{pmatrix} m & & \\ & mr^2 & \\ & & m \end{pmatrix},$$

$$T = \frac{m\dot{r}^2 + mr^2\dot{\varphi}^2 + m\dot{z}^2}{2}$$

Explicit expression for μ_{ab} in any curvilinear coordinates

By definition,

$$T = \frac{m}{2}\left(\frac{d\boldsymbol{r}}{dt}\right)^2$$

In general in orthogonal curvilinear coordinates (spherical, cylindrical, elliptical, parabolical, ...) we have

$$(d\boldsymbol{r})^2 = \sum_{a=1}^{3} h_a^2 \, (dr_a)^2$$

where h_a are called **Lamé coefficients,** may be found in any mathematical book or Wikipedia (don't confuse with Lamé constants/parameters for elastic materials, is completely different concept)

$$\Rightarrow \mu_{ab} = \delta_{ab} \ \left(m\, h_a^2\right),$$

$\Rightarrow$You don't need to evaluate μ_{ab} from scratch, may immediately write it using known h_a.

Lemma

$$0 = \frac{\delta \mathcal{F}}{\delta q_k} = \frac{\partial L}{\partial q_k} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) \quad (1)$$

If we make transformation of coordinates

$$q_a = q_a\left(r_1,\, r_2,\, r_3\right),$$

then the Euler-Lagrange equations in terms of variables r_a have a form

$$0 = \frac{\delta \mathcal{F}}{\delta q_k} = \frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right) = 0, \quad (2)$$

i.e. the Euler-Lagrange equations are *covariant*.

Proof:

$$\frac{\partial L}{\partial q_k} = \sum_{a=1}^{3} \frac{\partial r_a}{\partial q_k} \frac{\partial L}{\partial r_a},$$

$$\frac{\partial L}{\partial \dot{q}_k} = \sum_{a=1}^{3} \frac{\partial r_a}{\partial q_k} \frac{\partial L}{\partial \dot{r}_a}, \quad \dot{q}_k = \frac{\partial q_k}{\partial r_a} \dot{r}_a$$

$$\frac{\partial L}{\partial q_k} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) = \sum_{a=1}^{3} \frac{\partial r_a}{\partial q_k} \left[\frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right)\right] = 0$$

If $\det\left(\frac{\partial r_a}{\partial q_k}\right) \neq 0$, then equations

$$0 = \frac{\delta \mathcal{F}}{\delta q_k} = \frac{\partial L}{\partial q_k} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_k}\right) \quad (1)$$

$$0 = \frac{\delta \mathcal{F}}{\delta r_a} = \frac{\partial L}{\partial r_a} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{r}_a}\right) = 0 \quad (2)$$

are equivalent.

Lagrangian formalism

Equivalence of Euler-Lagrange equations in different frames ("*covariance*") means that we can write out the Lagragian and derive Euler-Lagrange equations in *any* system (not only Cartesian!).

The variables q_a in Euler-Lagrange equations are called "generalized coordinates", and the derivatives

$$p_k \equiv \frac{\partial L}{\partial \dot{q}_k}$$

are called "generalized momenta".

Note that with Euler-Lagrange approach we do not need to write out forces, make their projections etc. This is a powerful approach, and below we will see how it helps to integrate equations of motion for various systems

Generalized momenta - curvilinear coordinates

... the derivatives

$$p_k \equiv \frac{\partial L}{\partial \dot{q}_k}$$

are called "generalized momenta".
Examples:
▷ In Cartesian coordinates for free particles $p_i = m\dot{q}_i$
▷ In spherical coordinates for the same free particles we have

$$L = T = \sum_{a,b=1}^{3} \frac{\mu_{ab}(r)\,\dot{r}_a\dot{r}_b}{2} = m\frac{(\dot{r})^2 + r^2(\dot{\theta})^2 + r^2\sin^2\theta\,(\dot{\phi})^2}{2}$$

so $p_r = m\,\dot{r}$, $p_\theta = m\,r^2\dot{\theta}$, $p_\phi = m\,r^2\sin^2\theta\dot{\phi}$-can see that p_i is not just derivative of coordinate, might include extra prefactors

Mini-Summary

Covariance of Euler-Lagrange equations means that we can write out the Lagragian and derive Euler-Lagrange equations in *any* system (not only Cartesian!). We do not need to write out forces, make their projections etc. This is one of the important properties of the approach which makes it so useful.

In what follows we'll use a term "***generalized coordinates***" for *any* set of variables $\{q_a\}$ which permit to characterize the position of all the particles in the system (without specifying if it is Cartesian, spherical, elliptical ...) We'll call "***generalized momenta***" and "***generalized forces***" the variables

$$p_k \equiv \frac{\partial L}{\partial \dot{q}_k}, \qquad f_k = \frac{\partial L}{\partial q_k}$$

so the Euler-Lagrange equation gets the form similar to 2nd law of Newton:

$$\dot{p}_k = \frac{\partial L}{\partial q_k}.$$

Integrals of motion & Noether's theorem

- EL equations are **the second order ODEs**, might be difficult to integrate directly
- Some systems of ODEs possess first integrals:
  - After some manipulations, ODEs can be reducible to a form

$$\frac{d}{dt} Q\left(q_a,\, \dot{q}_b,\, t\right) = 0 \qquad (1)$$

    so clearly along the classical trajectory

$$Q\left(q_a,\, \dot{q}_b, t\right) = \mathrm{const} \qquad (2)$$

    where the value of const may be fixed from initial condition
  - Since (2) is the 1st order ODE, Mathematically it is much simpler than 2nd order Euler-Lagrange equation
  - Existence of the integrals of motion is closely related to symmetries of the system

Noether's theorem

For any continuous symmetry group of the action $\mathcal{S}$, there is a conserved quantity (integral of motion).

- Symmetry group of the action=set of transformations which do not change S (might be translations, rotations, ...).
  - *Symmetry group must be continuous. If transformation is discrete (NOT continuous), Noether theorem is not applicable.
- The Noether's theorem is valid in all branches of physics, always continuous symmetry corresponds to some integral of motion. We'll prove it only for classical mechanics below, but it might be extended (generalized) to other systems.

Integrals of motion & Noether's theorem

Noether's theorem

For any continuous symmetry of the action $\mathcal{S}$, there is a conserved quantity (integral of motion).

Examples:

- Invariance of L with respect to translations along $\boldsymbol{n}$

$$\boldsymbol{q} \to \boldsymbol{q} + \lambda\, \boldsymbol{n}, \quad \forall \lambda$$

  $\Rightarrow$conservation of momentum component $p_n = \boldsymbol{p} \cdot \boldsymbol{n}$

- Invariance of L with respect to rotations around $\hat{n}$ $\Rightarrow$conservation of angular momentum projection

$$M \cdot n \equiv n \cdot [\boldsymbol{q} \times \boldsymbol{p}]$$

- Time independence of L $\Rightarrow$conservation of energy

$$E \equiv \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} - L = T + U = \text{const}$$

Sometimes symmetries might be more complicated (e.g. potential $U(x - Vt)$ has symmetry with respect to $t \to t + t_0$, $x \to x + V\, t_0$). We'll learn a bit later how to construct invariants in this case.

Noether's theorem (proof)

Proof:

- Assume that $\epsilon = \mathrm{const}$ is a parameter of continuous symmetry transformation (translation, rotation, ...) which leaves the action S invariant
- Let's replace now $\epsilon = \mathrm{const}$ with $\epsilon = \epsilon(t) = \epsilon_0 + \delta\epsilon(t)$ with $\delta\epsilon\left(t_{a,b}\right) = 0$,. and $\delta\epsilon \ll \epsilon_0$
- First, let's assume that the transformation does not affect time, but changes the coordinates and velocities as:

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t), \qquad \dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t) + \dot{\epsilon}\gamma_a(t).$$

The classical trajectory is (local) minimum of the action, so small variations should NOT change S, since $\delta S/\delta q|_{q=q_{\mathrm{cl}}} = 0$. So we conclude that on classical trajectory $\delta S/\delta\epsilon(t) = 0$:

$$\delta S = \int_{t_1}^{t_2} dt\, \delta\epsilon \left(\frac{\partial L}{\partial \epsilon} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\epsilon}} \right) \right) = 0 \qquad (1)$$

Noether's theorem (II)

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t), \qquad \dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t) + \dot{\epsilon}\gamma_a(t).$$

The classical trajectory is (local) minimum of the action, so small variations should NOT change S, since $\delta S/\delta q|_{q=q_{\rm cl}} = 0$. So we conclude that on classical trajectory $\delta S/\delta\epsilon(t) = 0$:

$$\delta S = \int_{t_1}^{t_2} dt\, \delta\epsilon \left(\frac{\partial L}{\partial \epsilon} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\epsilon}}\right)\right) = 0 \qquad (1)$$

–The action S is invariant for transformations with $\epsilon = \mathrm{const}$, for this reason the lagrangian should be also invariant,

$$\frac{\partial L}{\partial \epsilon} \equiv \sum_a \left(\frac{\partial L}{\partial q_a}\gamma_a + \frac{\partial L}{\partial \dot{q}_a}\gamma_a \right) = 0 \qquad (2)$$

$$(1,2) \Rightarrow \frac{\partial L}{\partial \dot{\epsilon}} \equiv \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a = Q = \mathrm{const}$$

▷We obtained explicitly the Noether invariant Q.

Noether's theorem: applications

Noether's theorem

Let's assume that the lagrangian of the system L with n degrees of freedom is invariant with respect to

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t),$$

$$\dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t),$$

$$t \to t$$

where $\epsilon = \mathrm{const}$ is a continuous parameter. Then the quantity

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a$$

is conserved.

Applications (examples):

►If the action's symmetry group corresponds to translations along the axis $\boldsymbol{n}$, then clearly $\gamma_a \equiv n_a$, so the associated integral of motion is

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a = p \cdot n$$

-projection of momentum on $\boldsymbol{n}$

►If the action's symmetry group corresponds to rotations around $\boldsymbol{n}$, then

$$\delta\boldsymbol{q} = \delta\phi\, \boldsymbol{n} \times \boldsymbol{q} \Rightarrow \boldsymbol{\gamma} = \boldsymbol{n} \times \boldsymbol{q}$$

so the associated integral of motion is

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a = \boldsymbol{p} \cdot [\boldsymbol{n} \times \boldsymbol{q}] =$$

$$= \boldsymbol{n} \cdot [\boldsymbol{q} \times \boldsymbol{p}] = \boldsymbol{n} \cdot \boldsymbol{M}$$

-projection of angular momentum $\boldsymbol{M}$ on the direction of $\boldsymbol{n}$

Noether theorem and symmetries

Noether's theorem

Let's assume that the lagrangian of the system L with n degrees of freedom is invariant with respect to

$$q_a(t) \to q_a(t) + \epsilon\gamma_a(t),$$

$$\dot{q}_a(t) \to \dot{q}_a(t) + \epsilon\dot{\gamma}_a(t),$$

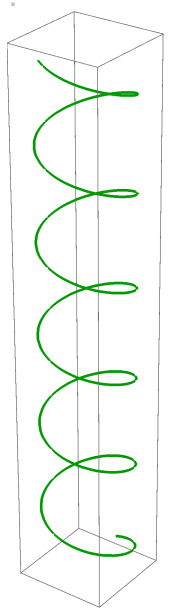
$$t \to t$$

where ϵ is a continuous parameter. Then the quantity

$$Q = \sum_a \frac{\partial L}{\partial \dot{q}_a}\gamma_a$$

is conserved.

Another example:



In cylindric coordinates potential $U = U(r, z - a\phi)$ has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ \phi \to \phi + \frac{\epsilon}{a} \quad (1)$$

▷Since kinetic energy is invariant w.r.t. rotations and translations, transformation (1) is symmerty group of the action

▷ Components of the γ-vector:

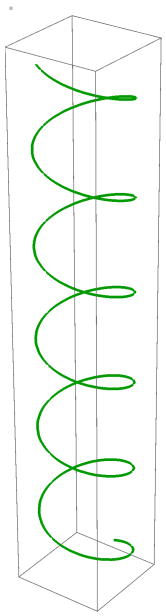
$$\gamma_r = 0,\ \gamma_\phi = \frac{\partial\phi}{\partial\epsilon} = a^{-1},\ \gamma_z = \frac{\partial z}{\partial\epsilon} = 1,$$

so the conserving quantity is

$$Q = \underbrace{\frac{\partial L}{\partial\dot{\phi}}}_{p_\phi \equiv M_z}\gamma_\phi + \underbrace{\frac{\partial L}{\partial\dot{z}}}_{p_z}\gamma_z = \frac{M_z}{a} + p_z$$

-neither p_z nor M_z conserve, yet their linear combination is constant (!)

Beware: only symmetries of S matter



▷ Action of a particle in external field:

$$S = \int dt\, L = \int dt \left[\frac{mv^2}{2} - U \right]$$

$$v^2 = g_{ij} v^i v^j$$

Compare 2 problems:

Assume that a charged particle moves in the external potential, which in cylindric coordinates is given by $U = U(r, z - a\phi)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ \phi \to \phi + \frac{\epsilon}{a} \quad (1)$$

Metrics in cylindr. coords: $g_{ij} = \{1, r^2, 1\}$ is invariant w.r.t. (1)
⇒There is integral of motion (Noether invariant) associated with symmerty (1) (which might be found taking derivative $\partial L / \partial \dot{\epsilon}$)

Assume that a charged particle moves in the external potential, which in cylindric coordinates is given by $U = U(z - a\,r)$. The potential has a continuous symmetry w.r.t.

$$z \to z + \epsilon,\ r \to r + \frac{\epsilon}{a} \quad (2)$$

Metrics $g_{ij} = \{1, r^2, 1\}$ is NOT invariant w.r.t. (2)
⇒There is NO integrals of motion (Noether invariants) associated with symmerty (2)

- Our previous derivation was based on the fact that symmetry transformation does not affect (change) time. Now we'll try to see example when time is also changed. We'll start for simplicity with the simplest symmetry transformation

$$q_a(t) \to q_a(t), \quad \dot{q}_a(t) \to \dot{q}_a(t), \quad t \to t + \epsilon \qquad (1)$$

Noether's theorem

If the lagrangian L of the system is invariant with respect to transformation (1), where ϵ is a continuous parameter, then there is a conserved quantity (energy)

$$E = \sum_a \frac{\partial L}{\partial \dot{q}_a} \dot{q}_a - L = \mathrm{const}$$

Elementary proof:

$$\frac{dL}{dt} = \frac{\partial L}{\partial t} + \sum_a \left(\frac{\partial L}{\partial q_a} \dot{q}_a + \frac{\partial L}{\partial \dot{q}_a} \ddot{q}_a \right) = \frac{\partial L}{\partial t} + \frac{d}{dt} \sum_a \left(\frac{\partial L}{\partial \dot{q}_a} \dot{q}_a \right)$$

$$\Rightarrow \frac{dE}{dt} = \frac{d}{dt} \sum_a \left(\frac{\partial L}{\partial \dot{q}_a} \dot{q}_a - L \right) = -\frac{\partial L}{\partial t}$$

Since by definition S is invariant under transformation (1), i.e. L does not depend explicitly on time, $\partial L / \partial t = 0$ and thus $E = \mathrm{const}$.

Alternative proof:

- Let's replace now $\epsilon = \mathrm{const}$ with $\epsilon = \epsilon(t) = \epsilon_0 + \delta\epsilon(t)$ with $\delta\epsilon\left(t_{a,b}\right) = 0$,. and $\delta\epsilon \ll \epsilon_0$

- The transformation does not affect coordinates, but changes the time and velocities as:

$$t \to t + \epsilon(t), \quad dt \to dt\left(1 + \dot{\epsilon}(t)\right), \quad \dot{q}_a \equiv \frac{dq}{dt} \to \frac{\dot{q}_a}{1 + \dot{\epsilon}(t)} \approx \dot{q}_a\left(1 - \dot{\epsilon}(t)\right) + \mathcal{O}\left(\epsilon^2\right)$$

Change of action under this transformation:

$$S \equiv \int dt\, L(q, \dot{q}) \to \int dt\, \left(1 + \dot{\epsilon}(t)\right) L(q, \dot{q}(1 - \dot{\epsilon})) \approx \tag{1}$$

$$\approx \int dt\, L(q, \dot{q}) + \int dt\, \dot{\epsilon}\left(L(q, \dot{q}) - \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a}\right) + \mathcal{O}\left(\epsilon^2\right)$$

$$S \approx \int dt\, L(q,\dot{q}) + \int dt\, \dot{\epsilon} \left(L(q,\dot{q}) - \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} \right) + \mathcal{O}\left(\epsilon^2\right)$$

- As in previous derivation, we consider that on classical trajectory the variational derivative should vanish,

$$\left.\frac{\delta S}{\delta q}\right|_{q=q_{\mathrm{cl}}} = \frac{\partial L}{\partial \epsilon} - \frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\epsilon}}\right) = 0 \qquad (2)$$

and $\partial L/\partial \epsilon = 0$ since for $\epsilon = \mathrm{const}$ the action S is invariant.

$$(1,2) \Rightarrow \frac{\partial L}{\partial \dot{\epsilon}} = L(q,\dot{q}) - \sum_a \dot{q}_a \frac{\partial L}{\partial \dot{q}_a} \equiv -E = \mathrm{const} \qquad \mathrm{Q.E.D}$$

Now can generalize this result for arbitrary symmetry transformation which mixes coordinates and time:

$$q \to q_\epsilon, \quad t \to t_\epsilon, \qquad \epsilon \ll 1$$

1) Make Taylor expansion of S, picking up the linear $\mathcal{O}(\epsilon)$, $\mathcal{O}(\dot{\epsilon})$ terms
2) All the $\sim \mathcal{O}(\epsilon)$ terms will vanish since for $\epsilon = \mathrm{const}$ the action is invariant
3) All the linear terms $\sim \mathcal{O}(\dot{\epsilon})$ allow to find the desired Noether invariant

$$Q = \lim_{\varepsilon \to 0} \frac{\delta S\left(q_\epsilon, \dot{q}_\epsilon, t_\epsilon\right)}{\delta \dot{\epsilon}} = \mathrm{const}$$

4) General result for invariant Q:

$$S_\epsilon = \int dt_\epsilon \, L\left(q_\epsilon, \dot{q}_\epsilon, t\right) = \int dt \left(\frac{\partial t_\epsilon}{\partial t}\right) L\left(q_\epsilon, \dot{q}_\epsilon, t\right)$$

$$\Rightarrow Q = \lim_{\varepsilon \to 0} \frac{\partial L_\epsilon\left(q_\epsilon, \dot{q}_\epsilon, t\right)}{\partial \dot{\epsilon}}, \quad L_\epsilon\left(q_\epsilon, \dot{q}_\epsilon, t\right) = \left(\frac{\partial t_\epsilon}{\partial t}\right) L\left(q_\epsilon, \dot{q}_\epsilon, t\right)$$

Example

Assume that the particle's lagrangian is given by

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} - U(x, y, z - V\,t)$$

Identify symmetry group and construct the integral of motion. Explain your result.

- Symmerty transform (Cartesian):

$$z \to z + V\,\epsilon,\; t \to t + \epsilon, \qquad \epsilon = \mathrm{const}$$

- Consider $\epsilon = \epsilon(t)$

$$dt \to dt\,(1 + \dot{\epsilon}(t))$$

$$\dot{\boldsymbol{r}} \to \frac{\dot{\boldsymbol{r}} + V\,\dot{\epsilon}\hat{\boldsymbol{z}}}{1 + \dot{\epsilon}} \approx \dot{\boldsymbol{r}}(1 - \dot{\epsilon}) + V\,\dot{\epsilon}\hat{\boldsymbol{z}} + \mathcal{O}\left(\dot{\epsilon}^2\right)$$

Change of lagrangian

$$L \to L_\epsilon = L\,(1 + \dot{\epsilon}) - \dot{\epsilon}\dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\dot{\epsilon}\hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot{z}} + \mathcal{O}\left(\dot{\epsilon}^2\right)$$

Example

$$L \to L_\epsilon = L\,(1+\dot\epsilon) - \dot\epsilon \dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\dot\epsilon \hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot z} + \mathcal{O}\left(\dot\epsilon^2\right)$$

▷ Construct Noether invariant:

$$\frac{\partial L}{\partial \dot\epsilon} = L - \dot{\boldsymbol{r}}_a \frac{\partial L}{\partial \dot{\boldsymbol{r}}_a} + V\,\hat{\boldsymbol{z}}\,\frac{\partial L}{\partial \dot z} = -E + p_z(t)V \quad (1)$$

Recall Galileos' transform for velocities and transformation of kinetic energy:

$$\frac{m\left(\boldsymbol{v}(t)-\boldsymbol{V}\right)^2}{2} = \frac{m\boldsymbol{v}^2(t)}{2} - \boldsymbol{p}(t)\cdot\boldsymbol{V} + \frac{mV^2}{2}$$

$V = \mathrm{const},\ v = v(t) = \dot r(t) \neq \mathrm{const}$

► Can rewrite (1) as

$$\frac{m\left(\boldsymbol{v}(t)-\boldsymbol{V}\right)^2}{2} = E_{\mathrm{rf}} + \frac{mV^2}{2}$$

where E_{rf} is the energy in the frame moving with velocity $-V\,\hat z$. In that frame

$$L = \frac{m\,\dot{\boldsymbol{r}}^2}{2} - U(x,y,z) + \frac{d}{dt}(\ldots)$$

does not depend on time, so energy is conserved

▷in contrast, in our frame potential depends on time, so $E \neq \mathrm{const}$.

Example 2

Demonstrate that the action S of a free particle is invariant with respect to rescaling transformation

$$\boldsymbol{r} \to \alpha\, \boldsymbol{r}, \quad t \to \alpha^2 t.$$

Construct the integral of motion which corresponds to this symmetry according to Noether theorem. Demonstrate that this quantity is conserving using the equations of motion.

Help: Instead of variable α You may find easier to work with variable $\lambda \equiv \ln \alpha$ (so $\alpha = e^{\lambda} \approx 1 + \lambda$ for $|\lambda| \ll 1$).

Example 2 (proof of invariance)

$$S = \int dt \, \frac{m \, \dot{\boldsymbol{r}}^2}{2}$$

Change of different quantities under rescaling:

$$\boldsymbol{r} \to \alpha \, \boldsymbol{r}, \;\; t \to \alpha^2 t, \qquad \alpha = e^{\lambda} \approx 1 + \lambda \qquad (1)$$

$$\dot{\boldsymbol{r}} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{1}{\alpha} \dot{\boldsymbol{r}} \approx \dot{\boldsymbol{r}} - \lambda \, \dot{\boldsymbol{r}},$$

$$\frac{m \, \dot{\boldsymbol{r}}^2}{2} \to \frac{1}{\alpha^2} \frac{m \, \dot{\boldsymbol{r}}^2}{2} \approx \frac{m \, \dot{\boldsymbol{r}}^2}{2} - 2\lambda \, \frac{m \, \dot{\boldsymbol{r}}^2}{2}$$

$$\Rightarrow S = \int dt \, L \to \int \cancel{\alpha^2} dt \, \frac{1}{\cancel{\alpha^2}} \frac{m \, \dot{\boldsymbol{r}}^2}{2} = \mathrm{const}$$

This proves that (1) is indeed a continuous symmetry of the action.

Example 2 (derivation of invariant)

►consider $\alpha = \alpha(t) = e^{\lambda(t)}, \quad |\lambda| \ll 1$

$$d\boldsymbol{r} \to (d\boldsymbol{r} + \boldsymbol{r} d\lambda) = \left(d\boldsymbol{r} + \boldsymbol{r}\,\dot{\lambda} dt\right)$$

$$dt \to \left(1 + 2\dot{\lambda} t\right) dt$$

$$\boldsymbol{v} \equiv \frac{d\boldsymbol{r}}{dt} \to \frac{\boldsymbol{v} + \boldsymbol{r}\,\dot{\lambda}}{1 + 2\dot{\lambda} t} \approx \boldsymbol{v} + \dot{\lambda}\,(\boldsymbol{r} - 2t\,\boldsymbol{v})$$

$$S \equiv \int dt\, L \to \int dt\, L\, \left(1 + 2\dot{\lambda} t\right)$$

$$+ \int dt\, m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v})\,\dot{\lambda} =$$

$$= \int dt\, \left[m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL\right]\dot{\lambda}$$

$$Q = \frac{\delta S}{\delta \dot{\lambda}} = m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2tL =$$

$$= m\,\boldsymbol{v} \cdot (\boldsymbol{r} - 2t\,\boldsymbol{v}) + 2t\left(\frac{m\,\boldsymbol{v}^2}{2}\right)$$

$$= m\,\boldsymbol{v} \cdot \boldsymbol{r} + 2t\left(-\frac{m\,\boldsymbol{v}^2}{2}\right) =$$

$$= \boldsymbol{p} \cdot \boldsymbol{r} - 2E\,t$$

▷we know that energy $\boldsymbol{E}$ and $\boldsymbol{p}$ are conserved for a free particle, and $\boldsymbol{p} = m\boldsymbol{v}$, $\boldsymbol{r} = \boldsymbol{v}t + \boldsymbol{r}_0$, so $Q = \boldsymbol{p} \cdot \boldsymbol{r}_0$indeed must conserve.

▷More formal check that $Q = \mathrm{const}$:

$$\frac{dQ}{dt} = \dot{\boldsymbol{p}} \cdot \boldsymbol{r} + \boldsymbol{p} \cdot \boldsymbol{v} - 2E$$

$$= -\frac{\partial U}{\partial \boldsymbol{r}} \cdot \boldsymbol{r} + \boldsymbol{p} \cdot \boldsymbol{v} - 2E =$$

$$= 2\left(\frac{m\,\boldsymbol{v}^2}{2}\right) - 2E \equiv 0$$

as expected !!!